N O U S

The Cognitive Intelligence Platform

5 Clinical Domains · 3 Intelligence Modules · 35 Patents · One Platform

NOST → NOSS → NOUS: From Theory to Platform

Version 8.0 | March 2026 | Beverly Index LLC
SCIENTIFIC FOUNDATION
🔬

Computational Psychodynamics

The Scientific Field

LEVEL 0 — SCIENTIFIC DISCIPLINE

The Science That Makes Everything Possible. Computational Psychodynamics is an entirely new scientific field that studies how to extract psychological information from what people say and write. Just like how chemistry studies molecules and physics studies energy, Computational Psychodynamics studies how your mind reveals itself through language.

In Simple Terms: Your words aren't random — they carry hidden patterns that reflect what's happening in your brain. Computational Psychodynamics is the science of finding and understanding these patterns. It's like having a special decoder that can read your mental state from the words you choose.

"Where Cognition Becomes Measurable"
Establishes the Scientific Framework ↓
🧠

NOST

Neural Oscillation Signature Theory

LEVEL 1 — FOUNDATIONAL THEORY

The Theory Behind Everything We Do. NOST is our foundational scientific theory within Computational Psychodynamics. It explains how your brain's electrical rhythms (neural oscillations) show up in the words you use and the way your voice sounds. Think of it as the "instruction manual" that tells all our tools how to understand what your language reveals about your mental state.

In Simple Terms: Your brain creates different electrical wave patterns depending on what you're thinking about or feeling. When you remember something real, your brain waves look different than when you're making something up. NOST discovered that these same patterns show up in how people talk and write. So by analyzing your words and voice, we can understand your brain patterns — whether you're recalling a real memory, processing trauma, or experiencing a mental health condition.

📊 NOSS Classification System

The Neural Oscillatory Spectroscopy System — our "periodic table" of mental states. Organizes 12 sectors of conditions by their oscillatory signatures, from Mood, Anxiety, and Psychotic Spectrum through Neurocognitive, Developmental, Optimal Function, TBI, Neurodegenerative, Captivity/Coercive Trauma, Substance Use, Dissociative, and Personality disorders. Each sector scored continuously $a = (a_1, a_2, \ldots, a_{12}) \in [0, 100]^{12}$, enabling multi-label classification.

🔄 Integration Dynamics

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Psychological Lense-Thirring Precession — tracks how your brain processes and integrates significant experiences through "wobble" patterns in dreams over time. Six pattern types from healthy integration to dissociative trajectories.

"The Science of Cognitive Signatures"
OPERATIONAL TECHNOLOGY
Powers the Technology Stack Below ↓
🛰

Adaptive Assessment Engine™

Intelligent Orchestration Layer

LEVEL 2 — USER-FACING FRONT DOOR

The System That Knows What It's Looking At. The AAE is the only component users interact with directly. You share whatever is on your mind — a dream, a memory, a conflict — and the AAE accepts it, analyzes it through REMIEL, determines what's happening, decides if it needs more information, and orchestrates the entire assessment pipeline automatically. No pre-selected packages, no human guessing — the system adapts to what the data reveals.

🕹

Universal Intake

Accepts ANY narrative input — dreams, conflicts, memories, speech — via text or voice

🔀

Intelligent Routing

Maps NOSS sector activation and routes to the appropriate assessment protocol automatically

🔄

Adaptive Deep-Dive

Detects data gaps and generates targeted follow-up prompts until clinical confidence is reached

Input: Your story (text or voice) → **Output:** Orchestrated assessment pipeline with full downstream processing

First Step: Establish Linguistic-Cultural Frame ↓
🌍

LOGOS™

Linguistic Origin & Gestural Oscillation System

LEVEL 2 — CROSS-CULTURAL CALIBRATION

The System That Knows Who Is Speaking. Before any analysis begins, LOGOS identifies the speaker's linguistic origin, dialect, cultural communication norms, and prosodic baseline. It builds a Linguistic-Cultural Frame (LCF) that calibrates every downstream component — REMIEL, POA, VOCA, the Beverly Index — against population-normed reference profiles appropriate to that specific speaker. Without LOGOS, NOUS measures deviation from an English-language norm. With LOGOS, NOUS measures deviation from the speaker's own cognitive-linguistic baseline.

🗣

6-Layer Architecture

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Language ID → Cultural Profiling → Prosodic Disentanglement → Code-Switch Analysis → Gestural Awareness → LCF Assembly

🎜

Tonal Disentanglement

Separates lexical tone from emotional prosody in Mandarin, Cantonese, Vietnamese, and 300+ tonal languages

📊

Cultural Expressiveness Profiles

Population-normed baselines ensure an Italian's passion isn't scored as mania and Finnish restraint isn't scored as flat affect

Input: Speaker's text and/or audio → **Output:** Linguistic-Cultural Frame (LCF) calibrating all downstream analysis

LCF Calibrates All Analysis — Sends Narratives to REMIEL ↓
📥

REMIEL Engine

Multi-Channel Analysis System

LEVEL 3 — 8-CHANNEL ANALYSIS

Reads Your Words and Voice Through Multiple Channels. REMIEL receives narratives from AAE and analyzes the language and vocal patterns — which words you use, how you describe things, what emotions come through, the structure of your narrative, and for spoken input, how your voice sounds. It extracts features across 8 input channels that reveal how your brain was working when you created those words.

💭 Dreams

Highest S/N ratio

✂️ Conflict

Attachment patterns

❤️ Emotional

Affect regulation

🔮 Future

Cognitive style

🪞 Self

Identity integration

🗣️ Speech

Real-time state

🎨 Creative

Thought flow

🎙️ POA

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Voice & prosody

Channel 8: Prosodic Oscillation Analysis (POA)

For spoken narratives, POA extracts acoustic-prosodic features across 6 sub-channels: pitch dynamics, temporal rhythm, energy contour, vocal quality, pause architecture, and articulation patterns. Enables Text-Voice Divergence Detection — identifying when what someone says contradicts how they say it.

From AAE: Tagged narrative → **Output:** Feature vectors across 8 channels + precession tracking for longitudinal analysis

Sends Feature Vectors to R2X ↓
🏭

R2X Unified Bridge

REMIEL-to-Everything Translation

LEVEL 4 — TRANSLATION ENGINE

🎯 What Does R2X Do?

The Translation Engine. R2X takes the feature vectors from REMIEL and uses NOST science to figure out what your brain was doing when you created those words. Then it generates raw visual parameters for DreamWeaver AND audio parameters for Memoria Sonata — all from a single analysis, perfectly synchronized. It also tracks precession dynamics for integration monitoring over time.

📥 → 🌍

Step 1: Receive

Gets feature vectors from REMIEL (all 8 channels)

🌍 → 📊

Step 2: Translate

Uses NOST + NOSS to find cognitive signatures

📊 → 🎨🎵

Step 3: Route

Generates parameters for visuals, audio, AND scoring

Parallel Output
🎨

DreamWeaver

2D Visual Output System

LEVEL 5a

Shows You What Your Mind Looks Like. DreamWeaver turns brain patterns into pictures — colorful 2D visualizations that represent your mental state. In forensic mode, it creates the NOST Signature Map showing cognitive patterns over time. Can stand alone or feed into Empyrean for full 3D synthesis.

From R2X: Visual parameters → **Output:** 2D pictures, maps, and integration visualizations

🎵

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Memoria Sonata

Audio Output System

LEVEL 5b

Lets You Hear Your Memories. Memoria Sonata turns brain patterns into sound — happy memories sound bright and warm, difficult memories might sound tense or distant. In forensic mode, it creates synchronized audio that plays alongside the visual analysis. Also sonifies integration progress over time.

From R2X: Audio parameters → **Output:** Sound you can hear

All Results Flow to Scoring System ↓
📊

Beverly Index Scoring System

Your Results Dashboard

LEVEL 6 — VERSION 2.0

The Final Score That Brings It All Together. The Beverly Index takes information from REMIEL's multi-channel analysis, DreamWeaver's visuals, and Memoria Sonata's audio — and produces easy-to-understand scores that help professionals make decisions. Version 2.0 includes integration dynamics tracking, expanded condition coverage including TBI and Parkinson's, and a 6-component scoring architecture.

📉 Diagnostic Probability

Likelihood scores for mental health conditions: MDD, GAD, PTSD, Bipolar, TBI categories (Concussion, PCS, CTE Risk), and Parkinson's prodromal markers

🎯 Diagnostic Coherence Index

How well all modalities agree (0.0-1.0) — cross-modal AND temporal coherence, including text-voice convergence for spoken input

🔄 Integration Dynamics Score

Tracks psychological integration efficiency via precession analysis — predicts healing timelines and treatment response

🏷 Integration Pattern

Categorical status (NORM-I, STUCK, DISORG, HYPER, AVOID, DISS-T) for rapid clinical triage and treatment planning

✅ Clinical Confidence Score

Overall actionability metric (0-100) combining probability, coherence, and integration dynamics

🔮 Authenticity Index

For CSAA pathway: How congruent a narrative is with authentic recall patterns (0-100 score)

Synthesized in 3D ↓
✨

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Empyrean™

3D Cognitive Visualization Synthesis

LEVEL 7 — VISUALIZATION LAYER

Your Mind, Mapped in 3D Space. Empyrean transforms all Beverly Index data into a navigable personal universe — a three-dimensional cognitive map unique to you. Just as astronomers map galaxies, Empyrean creates a visual representation of your mental landscape that you can explore, hear, and watch evolve over time. The name evokes the highest heaven — the clearest possible view.

🌌

Twelve Cognitive Sectors

NOSS sectors visualized as distinct 3D regions around a central Core

🌀

Core Spin Visualization

Your processing style shown as a rotating central Core with accretion disk

🛰️

Orbital Integration

Watch psychological material orbit and integrate with visible precession dynamics

The Twelve Sectors (Cosmic Metaphors)

🌟

S1 Mood

Stellar Nursery

☄️

S2 Anxiety

Asteroid Field

🌀

S3 Psychotic

Nebula

🛸

S4 Neurocog

Ancient System

⚛️

S5 Develop

Binary Star

🌍

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S6 Optimal

Goldilocks

💥

S7 TBI

Impact Crater

⏳

S8 Neurode

Dying Star

🔗

S9 Captivity

Event Horizon

🧪

S10 Substance

Rogue Orbit

🧭

S11 Dissociative

Parallel Dimension

♨️

S12 Personality

Magnetic Field

🎵 Synchronized Audio

Memoria Sonata provides spatial audio that changes as you navigate — hear different sectors sound different

⏱ Temporal Playback

Watch your Empyrean evolve over weeks/months — animate healing, see integration progress

From Beverly Index: All scores + integration data → **Output:** Interactive 3D cognitive map with synchronized audio

ADVANCED CAPABILITIES

🔊

VOCA

Vocal Oscillation Cognitive Archive

CONTINUOUS MONITORING LAYER

Real-Time Cognitive Monitoring Through Voice. VOCA extends the ecosystem beyond session-based assessment into continuous monitoring. It stores and learns each individual's unique vocal baseline over time, then detects meaningful deviations from *their* normal — not population averages. When your voice changes in ways that suggest depression onset, anxiety escalation, or

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neurodegenerative progression, VOCA detects it and alerts the appropriate practitioner.

🎙

Vocal Profile Storage

Encrypted storage of individual trait and state vocal features

📐

Adaptive Baselines

Dual-rate learning adapts to normal drift without normalizing pathology

⚡

Real-Time Detection

Sub-second deviation analysis on streaming audio with multi-dimensional scoring

🔔

Cognitive Alerts

Tiered alert system from informational to urgent based on deviation severity

Builds on: POA (Channel 8) feature extraction → **Adds:** Persistent individual baselines, adaptive learning, real-time streaming analysis, clinical alerts

CLINICAL SURVEILLANCE EXTENSION
Extends NOUS into Longitudinal Medical Surveillance — 5 Clinical Domains ↓
📐

Longitudinal Trajectory Engine

Per-Patient Time-Series Intelligence

SURVEILLANCE CORE — CHANGE-POINT DETECTION

From Snapshot to Surveillance. The Trajectory Engine transforms NOUS from a single-session assessment tool into a population-scale surveillance system. It maintains per-patient time-series databases across months and years, using Bayesian online changepoint detection to identify the exact moment when a cognitive-linguistic trajectory deviates from its established baseline — flagging the inflection point that may signal emerging cerebrovascular disease, cardiac pathology, or treatment-related neurotoxicity.

🗄

Per-Patient Time-Series

Every NOSS vector, oscillatory profile, and raw feature value stored longitudinally — designed for 36-48 month monitoring

📐

Bayesian Changepoint Detection

Statistical inflection-point identification — pinpoints when trajectory deviates from baseline with confidence scoring

🔔

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Tiered Alert System

GREEN (Routine) → YELLOW (Watch) → ORANGE (Evaluate) → RED (Urgent) — confidence-gated, actionable referrals

Input: Longitudinal NOSS vectors + EHR context → **Output:** Trajectory analysis, changepoints, differential classification, referral recommendations

Five Clinical Domains

🧠

Cerebrovascular

Vascular Cognitive Impairment

DOMAIN 2 — PAPER 4

Executive Decline Without Prosodic Change. Detects cerebrovascular disease through executive-predominant S4 trajectory inflection — processing speed, word-finding, syntactic complexity — WITHOUT prosodic-autonomic features. This absence of voice changes is the key differentiator from cardiac-origin cognitive decline.

Key Differentiator

Executive-predominant decline + NORMAL prosodic function = cerebrovascular, not cardiac. Correlates with undertreated hypertension, diabetes, hyperlipidemia.

Referral: Neurology — Brain MRI (white matter protocol), carotid duplex, coagulation panel

❤

Cardiac

Prosodic-Autonomic Detection

DOMAIN 3 — PAPER 5

Voice Changes Precede Cognitive Decline by Months. The architecture's core innovation. The Prosodic-Autonomic Analyzer detects cardiac pathology through F0 variability, speech rhythm, and respiratory patterning changes that appear months BEFORE executive cognitive symptoms. The Cardiac Pathway Discriminator then classifies across 5 cardiac pathways — each routing to a different specialist workup.

5 Cardiac Pathways

1. Reduced Output · 2. Microembolic Burden (AF) · 3. Autonomic Dysregulation · 4. Systemic Inflammation · 5. Shared Atherosclerosis

Referral: Cardiology — pathway-specific workup: echo, event monitor, autonomic testing, inflammatory panel, or CVD risk assessment

🎗

Oncology

Cancer Care Continuum

DOMAIN 4 — PAPER 6

From Chemo Brain to Immunotherapy Emergencies. Three specialized modules: the Cancer Care Continuum Tracker (4-phase monitoring from baseline to survivorship), the Treatment-Cycle Neurotoxicity Monitor (detecting cycle-synchronized cognitive perturbation — a pathognomonic pattern), and the Immunotherapy Safety Sentinel (high-sensitivity surveillance for the 1-5% of patients

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who develop neurological irAEs that are potentially fatal but REVERSIBLE if caught early).

6 Oncology Pathway Templates

1. Pre-treatment Inflammation · 2. Chemo Neurotoxicity · 3. Radiation Injury · 4. Immunotherapy irAE · 5. Brain Metastases · 6. Endocrine Therapy

Referral: Neuro-Oncology / Neuroimmunology — cycle-specific dose modification, STAT MRI, antibody panels, IV steroid preparation

GENESIS™

Prenatal & Maternal-Fetal Health

DOMAIN 5 — PAPER 7

Maternal Language as a Window to Fetal Risk. GENESIS monitors cognitive-linguistic and prosodic-autonomic biomarkers in pregnant women to detect preeclampsia, gestational diabetes, perinatal mood disorders, and stillbirth risk factors. Maternal stress physiology encoded in vocal and linguistic features provides a non-invasive, continuous monitoring channel that complements traditional prenatal screening.

5 Prenatal Pathways

1. Preeclampsia Risk · 2. Gestational Diabetes · 3. Perinatal Mood Disorder · 4. Stillbirth Risk Factors · 5. Maternal Mortality Forecasting

Referral: Maternal-Fetal Medicine — enhanced monitoring protocol, high-risk OB escalation, mental health integration, postpartum surveillance

Differential Signature Engine

The System That Tells You WHY — Not Just WHAT

Probabilistic Classification, Not Labels. The Differential Signature Engine classifies trajectory changes against etiological reference templates — outputting a probability distribution across all possible causes, not a single diagnosis. It integrates EHR data (labs, imaging, biomarkers, medications) to refine classification confidence, and each classification routes to specific specialist referrals with recommended workup protocols.

Cerebrovascular

Executive-predominant, no prosodic

Cardiac

Prosody-first temporal sequence

Neurodegenerative

Memory-predominant, progressive

Psychiatric

Content-specific, S1/S2/S3

Normal Aging

Slow, diffuse, no inflection

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Prenatal/Maternal

Stress-autonomic, hormonal

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EHR Data Intake

Bidirectional FHIR Integration

Clinical Context Changes Everything. Ingests labs, imaging, cardiac monitoring, and medication data through FHIR API integration. Blood pressure, HbA1c, LDL, hs-CRP, BNP, troponin, carotid ultrasound, brain MRI reports, chemotherapy regimens, and cardiac rhythm data — all feeding into the Differential Engine to refine classification confidence.

Integration: FHIR R4 API • Bidirectional • Lab results • Imaging reports • Medication lists • Treatment schedules • Vital signs • Device data

📊

BLUF Dashboard

Bottom-Line-Up-Front Clinical Interface

One Color. One Sentence. Everything Else Below. The NOUS Dashboard delivers clinical intelligence using military-grade BLUF methodology: the clinician sees one severity-coded signal and one sentence. Six progressive disclosure layers beneath — Beverly Index scoring, NOSS Spectrum Map, NOSS Analysis, Empyrean visualization, Treatment Intelligence, and Clinical Surveillance Intelligence — only surface when pulled.

6 Pull Layers: Beverly Index → NOSS Map → NOSS Analysis → Empyrean → Treatment Intelligence → Surveillance Intelligence

💊

Treatment Intelligence

NOSS-Matched Treatment Recommendations

From Diagnosis to Action. Dual-panel treatment recommendation system that matches detected NOSS sector signatures to published treatment modality evidence. A local panel provides geo-matched facilities with distance and specialty. A worldwide panel surfaces advanced protocols, clinical trials, and emerging treatments without geographic restriction. Each recommendation carries an evidence classification (RCT Published, Evidence-Based, Gold Standard, Phase II/III, Emerging) and match percentage.

📍 Local Panel

Geo-matched facilities, distance, description, NOSS alignment score

🌍 Worldwide Panel

Advanced protocols, clinical trials, emerging treatments, facility + evidence level

INTELLIGENCE LAYER
Cross-Domain Analytical Modules — Consuming All Diagnostic Outputs ↓
⚡

Intelligence Layer

Higher-Order Predictive Analytics

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CROSS-DOMAIN — COMMERCIALLY MODULAR

Where Surveillance Becomes Foresight. The Intelligence Layer sits atop all five clinical domains and transforms diagnostic data into predictive, prescriptive, and certifying intelligence. Each module consumes NOSS outputs from the Diagnostic Layer and generates higher-order analytics that no single domain could produce alone. Modules are **commercially independent** — customers activate only what they need — but **architecturally integrated**, sharing data substrates that improve accuracy across the entire platform.

The Strategic Principle: NOUS is the platform. The modules are the products. The platform is the moat. Every analysis across every module feeds the same analytical engine. Every customer activation deepens the data substrate. Every new domain or module extends the value of every existing one.

⏳

THANATOS™

Mortality Forecasting Module

INTELLIGENCE MODULE 1

Predictive Life-Course Surveillance. THANATOS analyzes multi-domain NOSS decay trajectories to forecast mortality risk at 6, 12, and 36-month horizons. By tracking the rate and pattern of cognitive-linguistic deterioration across all twelve NOSS sectors simultaneously, it identifies convergent decline signatures that precede end-of-life transitions — often months before clinical indicators.

Core Constructs

Δ-NOSS Temporal Decay Analysis · Mortality Convergence Index (MCI) · Terminal Linguistic Marker Profile (TLMP) · 6/12/36-Month Risk Stratification

Markets: Life insurance & actuarial · Palliative & hospice care · Pharmaceutical clinical trials · Elder care monitoring

Year 3: $15–25M · Year 5: $68–100M

💊

ORACLE™

Pharmacological Response Prediction

INTELLIGENCE MODULE 2

Precision Medicine Through Language. ORACLE predicts individual drug efficacy, adverse reactions, and optimal dosing by analyzing pre-treatment cognitive-linguistic biomarker profiles. Rather than genomic testing alone, it captures the brain's functional state — how it's actually processing — to predict how it will respond to pharmacological intervention.

Core Constructs

Pharmacological Receptivity Index (PRI) · Adverse Response Signature (ARS) · Therapeutic Window Estimation (TWE) · Treatment Trajectory Modeling (TTM)

Markets: Pharma trial stratification · Clinical decision support (SaaS) · Health system enterprise licensing · Psychiatry & neurology practices

Year 3: $12–20M · Year 5: $47–70M

🛡

AEGIS™

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Cognitive Fitness Certification

INTELLIGENCE MODULE 3

On-Demand Cognitive Performance Verification. AEGIS provides non-invasive, real-time cognitive fitness certification for safety-critical roles. A 3–5 minute assessment produces a definitive fitness-for-duty determination — detecting fatigue, impairment, medication effects, and subclinical deterioration that traditional screening misses.

Core Constructs

Cognitive Performance Score (CPS) · Fitness-for-Duty Thresholds (FFD) · Impairment Detection Array (IDA) · Longitudinal Performance Baseline (LPB)

Markets: Defense & military (DoD) · Commercial aviation (FAA) · Healthcare systems (surgical) · Professional sports · Commercial fleet

Year 3: $18–30M · Year 5: $81–130M

🔄

Cross-Module Data Flywheel

Each Module Makes Every Other Module More Valuable

ORACLE → THANATOS

Treatment response data informs mortality trajectories — can pharmacological intervention alter a patient's predicted decline curve?

AEGIS → ORACLE

Real-time medication side-effect signatures feed back to ORACLE's Adverse Response Signature library at unprecedented scale

AEGIS → THANATOS

Longitudinal Performance Baselines from healthy populations create normative datasets that sharpen decay curve detection

THANATOS → ORACLE

Mortality risk scores prioritize which patients most urgently need optimized pharmacological selection

ORACLE → AEGIS

Flags medications likely to degrade CPS scores, enabling proactive fitness threshold adjustment for medicated personnel

GENESIS → ORACLE

Prenatal data extends ORACLE to maternal pharmacological safety — predicting fetal response through maternal linguistic biomarkers

Intelligence Layer Revenue Projection: Year 3: $45–75M · Year 5: $196–300M

Incremental to Diagnostic Layer revenue from Domains 1–5

INFRASTRUCTURE & COMPLIANCE
🔒

HIPAA / Security

AES-256 encryption at rest, TLS 1.3 in transit. PHI segmentation — linguistic data stored separately

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from identifiers, linked by tokens. Complete audit logging, role-based access, BAA framework, SOC 2 Type II compliance.

⚕

FDA SaMD Pathway

Software as a Medical Device — Class II 510(k) for clinical surveillance alert/recommendation system. Clinical validation studies (Papers 4, 5, 6) designed for FDA requirements. CE Marking for EU MDR 2017/745. The system recommends; the clinician decides.

☁

Cloud & Edge

Kubernetes microservices on HIPAA-compliant cloud. Edge processing for on-device audio feature extraction. Multi-tenant with complete data isolation. Designed for population-scale surveillance across thousands of concurrent patients.

🎯 Five Ways We Help

💬

Cognitive Intelligence

Mental Health Analysis

Understanding Your Mind Through Narratives. Via the AAE, we analyze dream and narrative input to help identify mental health conditions — faster and more objectively than traditional methods. Integration Dynamics tracking monitors therapeutic progress and predicts healing timelines.

Who uses it: Therapists, psychiatrists, mental health clinicians, sports medicine, military healthcare, neurologists

MDD GAD PTSD Bipolar Psychotic TBI/CTE Parkinson's ADHD/ASD Captivity Substance DID Personality
🔍

CSAA

Cognitive Signature Authenticity Assessment

Understanding If a Story Is Authentic. A separate forensic pathway that analyzes any narrative to determine if it reflects genuine memory recall or constructed fabrication — providing objective, scientific authenticity assessment. CSAA operates its own standardized forensic protocol independent of AAE.

Who uses it: Security agencies, insurance investigators, legal professionals, HR departments, immigration services

🎙

VOCA Monitoring

Continuous Cognitive Surveillance

Ongoing Monitoring Between Sessions. VOCA enables persistent tracking of cognitive state through learned vocal baselines. Detects depression onset, anxiety escalation, PTSD activation, Parkinson's progression, and treatment response in real-time — flagging changes before self-report.

Who uses it: Military readiness programs, telehealth providers, neurology clinics, POW/hostage

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recovery, treatment monitoring

🏥

Clinical Surveillance

Vascular · Cardiac · Oncology · Prenatal Extension

Detecting Medical Conditions Through Language — Months Before Symptoms Appear. Clinical Surveillance extends NOUS beyond mental health into vascular, cardiac, oncological, and prenatal monitoring. By tracking longitudinal trajectory changes in cognitive-linguistic features and prosodic-autonomic biomarkers, the system detects cerebrovascular disease, cardiac pathology, chemotherapy neurotoxicity, immunotherapy adverse events, and maternal-fetal risk factors — often months before traditional clinical presentation.

Who uses it: Cardiologists, neurologists, neuro-oncologists, maternal-fetal medicine, primary care (population health), oncology treatment teams, survivorship programs

Cerebrovascular Cardiac Chemo Neurotoxicity Immunotherapy irAE Radiation Injury Survivorship Preeclampsia Stillbirth Risk
⚡

Intelligence Layer

THANATOS · ORACLE · AEGIS — Predictive, Prescriptive, Certifying

From Understanding to Foresight. The Intelligence Layer transforms diagnostic outputs into actionable predictions. THANATOS forecasts mortality trajectories. ORACLE predicts pharmacological response. AEGIS certifies cognitive fitness for safety-critical roles. Each module is commercially independent but architecturally integrated — sharing data substrates that create compounding value.

Who uses it: Life insurers, hospice care, pharma trial sponsors, defense (DoD), commercial aviation (FAA), sports organizations (NFL/NBA), surgical teams, fleet operators

THANATOS™ ORACLE™ AEGIS™ Mortality Forecasting Drug Response Fitness-for-Duty
📋

NOST Signature Report

The CSAA Deliverable

Forensic-Grade Authenticity Documentation. The NOST Signature Report is what CSAA produces — a professional document that captures the "cognitive fingerprint" of a narrative and provides objective, quantifiable evidence of whether it reflects authentic recall or constructed fabrication.

Executive Summary

1 page — Quick assessment

Standard Report

3-5 pages — With visuals

Full Forensic

10+ pages — Legal ready

📋 The Complete Journey

1️⃣

You share a story

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A dream, a narrative, an account of events — by typing or speaking into the Adaptive Assessment Engine

2

AAE accepts and routes your input

The system determines what you've shared, identifies concern areas, and may ask targeted follow-ups to fill data gaps

3

LOGOS identifies who is speaking

Language, dialect, cultural expressiveness norms, tonal system — builds a Linguistic-Cultural Frame that calibrates all downstream analysis

4

REMIEL analyzes through 8 channels

Finding patterns in word choice, structure, emotion, narrative coherence — calibrated by LOGOS to the speaker's linguistic-cultural baseline

5

R2X translates to brain patterns

Using NOST + NOSS to figure out what your brain was doing — outputs to visuals, audio, AND tracks integration dynamics

6

DreamWeaver shows you visuals

A visual picture of your mental state — or a NOST Signature Map for authenticity analysis

7

Memoria Sonata plays your sound

So you can actually hear what your narrative "sounds like" — synchronized with the visuals

8

Beverly Index v2.0 combines everything

Diagnostic probabilities, coherence index, integration dynamics, pattern classification, authenticity score, and clinical confidence — the complete picture

9

Empyrean visualizes your mind in 3D

Navigate your personal cognitive universe — twelve sectors, orbital dynamics, Core Spin, synchronized audio, temporal playback

10

Longitudinal Trajectory Engine monitors over months and years

Per-patient time-series tracking with Bayesian changepoint detection — identifies the exact moment a trajectory deviates from baseline

11

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Differential Signature Engine classifies the cause

Cerebrovascular, cardiac, neurodegenerative, psychiatric, or normal aging — with EHR data integration for clinical context

1️⃣2️⃣

Referral Pathway Router connects you to the right specialist

Classification → specific specialist → recommended workup protocol → clinical rationale. The system recommends; the clinician decides.

1️⃣3️⃣

THANATOS™ forecasts mortality trajectories

Multi-domain NOSS decay analysis produces 6/12/36-month risk stratification — enabling proactive palliative planning and clinical trial enrollment

1️⃣4️⃣

ORACLE™ predicts pharmacological response

Pre-treatment cognitive-linguistic biomarkers predict drug efficacy, adverse reactions, and therapeutic windows — precision medicine through language

1️⃣5️⃣

AEGIS™ certifies cognitive fitness

3–5 minute on-demand assessment certifies fitness-for-duty in safety-critical roles — pilots, surgeons, soldiers, commercial drivers

🔑 The Key Insight

Computational Psychodynamics is the field. The science of understanding mind through language.
NOST is the theory. NOSS is the classification system. They form the framework that connects everything.
NOUS is the platform. The Cognitive Intelligence Platform that houses every technology below.
AAE is the front door. It accepts your input, routes intelligently, and orchestrates the entire assessment.
LOGOS is the cultural lens. It knows who is speaking and calibrates every measurement accordingly.
REMIEL is the reader. It analyzes your words and voice through 8 channels and extracts features.
R2X is the translator. It converts features to brain patterns for visuals, audio, AND integration tracking.
DreamWeaver shows. Memoria Sonata plays.
Beverly Index v2.0 scores. Probability + Coherence + Integration + Authenticity = Clinical Confidence.
Empyrean maps. Your mind visualized as a navigable 3D universe.
VOCA monitors. Continuous cognitive surveillance through learned vocal baselines.
Clinical Surveillance extends. Trajectory Engine + Differential Engine + Referral Router = vascular, cardiac, oncology, and prenatal detection months before symptoms.
GENESIS protects. Maternal-fetal surveillance through language — preeclampsia, stillbirth risk, perinatal mood disorders.
Treatment Intelligence acts. NOSS-matched recommendations connect detection to intervention.
Intelligence Layer predicts. THANATOS forecasts. ORACLE prescribes. AEGIS certifies. Cross-domain modules that transform diagnosis into foresight.

NOST → NOSS → NOUS: Theory → Classification → Platform

📐 Technology Hierarchy

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Level 0: Computational Psychodynamics (Scientific Field)

↓

Level 1: NOST (Theory) + NOSS (Classification System)

↓

N O U S — THE COGNITIVE INTELLIGENCE PLATFORM

Level 2: AAE™ (Orchestration) + LOGOS™ (Cross-Cultural Calibration)

↓

Level 3: REMIEL Engine (8-Channel Analysis incl. POA)

↓

Level 4: R2X Bridge (Translation & Parameter Generation)

↓

Level 5a: DreamWeaver

Level 5b: Memoria Sonata

↓

Level 6: Beverly Index v2.0 (6-Component Scoring Output)

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Level 7: Empyrean™ (3D Cognitive Visualization Synthesis)

+ Advanced Capabilities

VOCA (Continuous Monitoring)

CSAA (Forensic Pathway)

+ Clinical Surveillance Extension

Longitudinal Trajectory Engine (Per-Patient Time-Series + Changepoint Detection)

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Cerebrovascular Detection

Cardiac Detection (5 Pathways)

Oncology Extension (6 Templates)

GENESIS™ (Prenatal)

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Differential Signature Engine

Referral Pathway Router

+ Output & Intelligence

BLUF Dashboard

Treatment Intelligence

EHR Integration (FHIR)

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+ Intelligence Layer (Cross-Domain Modules)

THANATOS™ (Mortality Forecasting)

ORACLE™ (Pharma Response)

AEGIS™ (Cognitive Fitness)

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NOUS™ — The Cognitive Intelligence Platform

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